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The empty set (and friends)

The symbol that started it all:

$$\emptyset \quad \varnothing \quad \{x \in \mathbb{R} : x \neq x\} = \varnothing.$$

Compare in context:

$$A \cap B = \varnothing, \qquad \bigcap_{n \in \mathbb{N}} (0, 1/n) = \varnothing, \qquad \mathscr{P}(\varnothing) = \{\varnothing\}.$$

Number sets and blackboard bold

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}, \qquad \mathbb{F}_p, \quad \mathbb{H}, \quad \mathbb{P}^n, \quad \mathbb{1}_A.$$

Greek alphabet

Lowercase:

$$\alpha\,\beta\,\gamma\,\delta\,\epsilon\,\varepsilon\,\zeta\,\eta\,\theta\,\vartheta\,\iota\,\kappa\,\lambda\,\mu\,\nu\,\xi\,o\,\pi\,\varpi\,\rho\,\varrho\,\sigma\,\varsigma\,\tau\,\upsilon\,\phi\,\varphi\,\chi\,\psi\,\omega$$

Uppercase:

$$\Gamma\,\Delta\,\Theta\,\Lambda\,\Xi\,\Pi\,\Sigma\,\Upsilon\,\Phi\,\Psi\,\Omega$$

Calligraphic and script

$$\mathscr{A}\,\mathscr{B}\,\mathscr{C}\,\mathscr{D}\,\mathscr{E}\,\mathscr{F}\,\mathscr{G}\,\mathscr{H}\,\mathscr{I}\,\mathscr{J}\,\mathscr{K}\,\mathscr{L}\,\mathscr{M}\,\mathscr{N}\,\mathscr{O}\,\mathscr{P}\,\mathscr{Q}\,\mathscr{R}\,\mathscr{S}\,\mathscr{T}\,\mathscr{U}\,\mathscr{V}\,\mathscr{W}\,\mathscr{X}\,\mathscr{Y}\,\mathscr{Z}$$

Logic and set theory

$$\forall x \in X,\ \exists y \in Y \text{ such that } P(x, y) \implies Q(y).$$

$$A \cup B,\ A \cap B,\ A \setminus B,\ A \subset B,\ A \subseteq B,\ A \subsetneq B,\ A^\complement,\ \mathscr{P}(A).$$

$$\neg p \vee (q \wedge r) \iff (\neg p \vee q) \wedge (\neg p \vee r).$$

Relations

$$= \neq \approx \equiv \cong \sim \simeq \asymp \propto \leq \geq \ll \gg \prec \succ \preceq \succeq \perp \parallel .$$

Operators and big operators

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}, \qquad \prod_{p \leq N} \left(1 - \frac{1}{p^s}\right)^{-1} = \sum_{n=1}^{\infty} \frac{1}{n^s} = \zeta(s).$$

$$\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x = \sqrt{\pi}, \qquad \iint_D f(x, y)\,\mathrm{d}A, \qquad \oint_{\partial\Sigma} \vec{F} \cdot \mathrm{d}\vec{r}.$$

$$\bigcup_{n \in \mathbb{N}} A_n, \quad \bigcap_{n \in \mathbb{N}} A_n, \quad \bigoplus_{i \in I} V_i, \quad \bigotimes_{i=1}^{n} H_i, \quad \biguplus, \quad \bigsqcup.$$

Calculus

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_a^x f(t)\,\mathrm{d}t = f(x), \qquad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}, \qquad \nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}.$$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

Showcase equations

Cauchy–Schwarz:

$$|\langle u, v \rangle|^2 \leq \langle u, u \rangle \cdot \langle v, v \rangle.$$

Binomial theorem:

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

Euler's identity:

$$e^{i\pi} + 1 = 0.$$

Stokes' theorem:

$$\int_{\partial\Omega} \omega = \int_{\Omega} \mathrm{d}\omega.$$

Linear algebra

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \qquad \det \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

$$\langle Ax, y \rangle = \langle x, A^\top y \rangle, \qquad \|x\|_2 = \sqrt{\sum_i x_i^2}, \qquad \mathrm{tr}(AB) = \mathrm{tr}(BA).$$

Big delimiters and nesting

$$\left(\sum_{i=1}^{n} a_i b_i \right)^2 \leq \left(\sum_{i=1}^{n} a_i^2 \right) \left(\sum_{i=1}^{n} b_i^2 \right).$$

$$\left\{ x \in \mathbb{R}^n \,:\, \|x - \mu\|_2^2 \leq \chi^2_{\alpha,n} \right\}.$$

Fractions, roots, and continued fractions

$$\cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cdots}}} = \frac{\sqrt{5} - 1}{2}.$$

$$\sqrt[n]{x}, \quad \sqrt{a + \sqrt{b + \sqrt{c}}}, \quad \frac{a+b}{c-d}, \quad \binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}}.$$

Arrows

$$\to \rightarrow \leftarrow \leftrightarrow \Rightarrow \Leftarrow \Leftrightarrow \mapsto \hookrightarrow \twoheadrightarrow \rightharpoonup \rightleftharpoons \nearrow \searrow .$$

Miscellaneous symbols

$$\hbar,\ \ell,\ \Re,\ \Im,\ \wp,\ \aleph,\ \beth,\ \partial,\ \nabla,\ \infty,\ \angle,\ \triangle,\ \dagger,\ \ddagger,\ \star,\ \ast,\ \circ,\ \bullet.$$

Inline-heavy paragraph

For $x \in \mathbb{R}^n$ and $\varepsilon > 0$, the open ball $B_\varepsilon(x) = \{y \in \mathbb{R}^n : \|y - x\| < \varepsilon\}$ is a basic open set in the Euclidean topology. A function $f\colon X \to Y$ between topological spaces is continuous if $f^{-1}(U)$ is open in X for every open $U \subseteq Y$, equivalently $\forall x \in X,\ \forall \varepsilon > 0,\ \exists \delta > 0$ such that $d_X(x', x) < \delta \implies d_Y(f(x'), f(x)) < \varepsilon$. The empty set $\emptyset$ is open and closed in every topology, and $f^{-1}(\emptyset) = \emptyset$, $f(\emptyset) = \emptyset$.